Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Venous thromboembolism – obstetric prophylaxis and management
Eastern Maternity Services

Summary

This guideline outlines the assessment and management of antenatal, intrapartum and postnatal thromboprophylaxis. It also highlights investigation and management of proven thromboembolism in pregnancy

Key Points

- A thromboembolism assessment should be performed on all women, antenatally, intrapartum and postnatally.
- Women with risk factors may require thromboprophylaxis and the doses will be booking weight dependant.

Weight (Kg)	Standard Enoxaparin Dose	Standard Dalteparin Dose
<50	20mg daily	2500 units daily, S/C
50-90	40mg daily	5000 units daily, S/C
91-130	60mg daily *	7500 units daily, S/C
131-170	80mg daily *	10,000 units daily S/C
>170	0.6mg/kg/day *	75 units/kg/day
High prophylactic dose for 50-90kg	40mg twice daily	5000iu twice daily
*** may be given in 2 divided doses**		

- Any woman commenced on Low molecular weight heparin (LMWH) antenatally must have a clear management plan documented in the chart including the plan for delivery and a referral should be made to the anaesthetic team.
- Risk assessment information can be found in the VTE tab.

CONTENTS

Section		Page
1	Introduction	3
2	Pre-pregnancy and antenatal risk assessment	3
3	Management of women with thrombophilia associated VTE	6
4	Labour and delivery	8
5	Third stage of labour	9
6	Postnatal risk assessment and thromboprophylaxis	9
7	Diagnosis and Treatment of Suspected VTE	12
8	Postnatal follow up for patients with VTE in pregnancy and the puperium	16
9	Monitoring compliance with this guideline	17
10	References	17
11	Publication details	17

1.0 INTRODUCTION

Thrombosis and venous thromboembolism (VTE) were the leading cause of direct maternal death in the MBRRACE UK report.

2.0 PRE-PREGNANCY AND ANTENATAL RISK ASSESSMENT

2.1 Women at high risk of VTE in pregnancy (such as those with previous VTE), should be offered pre-pregnancy counseling in the pre-conceptual antenatal clinic and a prospective management plan for thromboprophylaxis in pregnancy made.

2.2 **Antenatal thromboprophylaxis should begin as early in pregnancy as practical.**

Antenatal risk assessment for VTE (via VTE assessment tab) should be completed at booking by the community midwife.

Risk assessment should be repeated and documented:

- at any admission to hospital
- if another intercurrent problem develops (such as the transient risk factors in chart below)
- At 28 weeks

The risk of VTE should be discussed with women at risk and reasons for individual recommendations explained.

2.3 Depending on their risk factors patients may require thromboprophylaxis from first trimester or 28 weeks.

2.4 Women with previous VTE should have a careful history documented. Where objective documentation is not available, the previous diagnosis of VTE can be assumed in cases where the woman gives a good history and received prolonged (>6 weeks) therapeutic anticoagulation.

2.5 **Antenatal Risk Assessment** (Adapted from RCOG Green top Guideline)

Any previous VTE except single event related to major surgery.

See diagram 2 for additional management of patients with thrombophilia

→

High Risk

Requires antenatal prophylaxis with LMWH

Liaise with GP and start dalteparin **ASAP**

Refer to antenatal clinic

HOSPITAL ADMISSION

Single previous VTE related to major surgery

High risk thrombophilia*+ no previous VTE (*Antiphospholipid syndrome/ antithrombin deficiency/ protein s or c deficiency/compound or homozygous for low risk thrombophilias i.e. factor v leiden)

MEDICAL COMORBITIES, e.g. Heart failure, active SLE, cancer, inflammatory bowel disease or inflammatory polyarthropathy, nephrotic syndrome, sickle cell disease.

Current intravenous drug user

Surgical procedure e.g. appendicectomy

Ovarian hyperstimulation (first trimester only)

→

Intermediate Risk

Consider antenatal prophylaxis with LMWH

Refer to antenatal clinic for assessment if not already admitted.

Age >35 years
BMI 30 - 39
BMI >40 (= 2 risk factors)
Parity ≥3
Smoker
Gross varicose veins
Current pre- eclampsia
Immobility e.g. paraplegia, PGP
Multiple pregnancy
Family history of unprovoked or oestrogen provoked VTE in 1st degree relative
Low risk thrombophilia* (*Heterozygous for factor V Leiden or prothrombin G20210A mutations)
IVF/Assisted reproduction
Transient risk factors Dehydration/ hyperemesis; current systemic infection; long distance travel.

→

4 or more risk factors: **Consider prophylaxis from first trimester**

3 risk factors: **Consider prophylaxis from 28 weeks**

Fewer than 3 risk factors

→

Lower Risk

Mobilisation and avoidance of dehydration

2.6 Antenatal Thromboprophylaxis

Low molecular weight heparins (LMWH) are the agents of choice for antenatal thromboprophylaxis. The dose of LMWH depends on the booking weight (table 1).

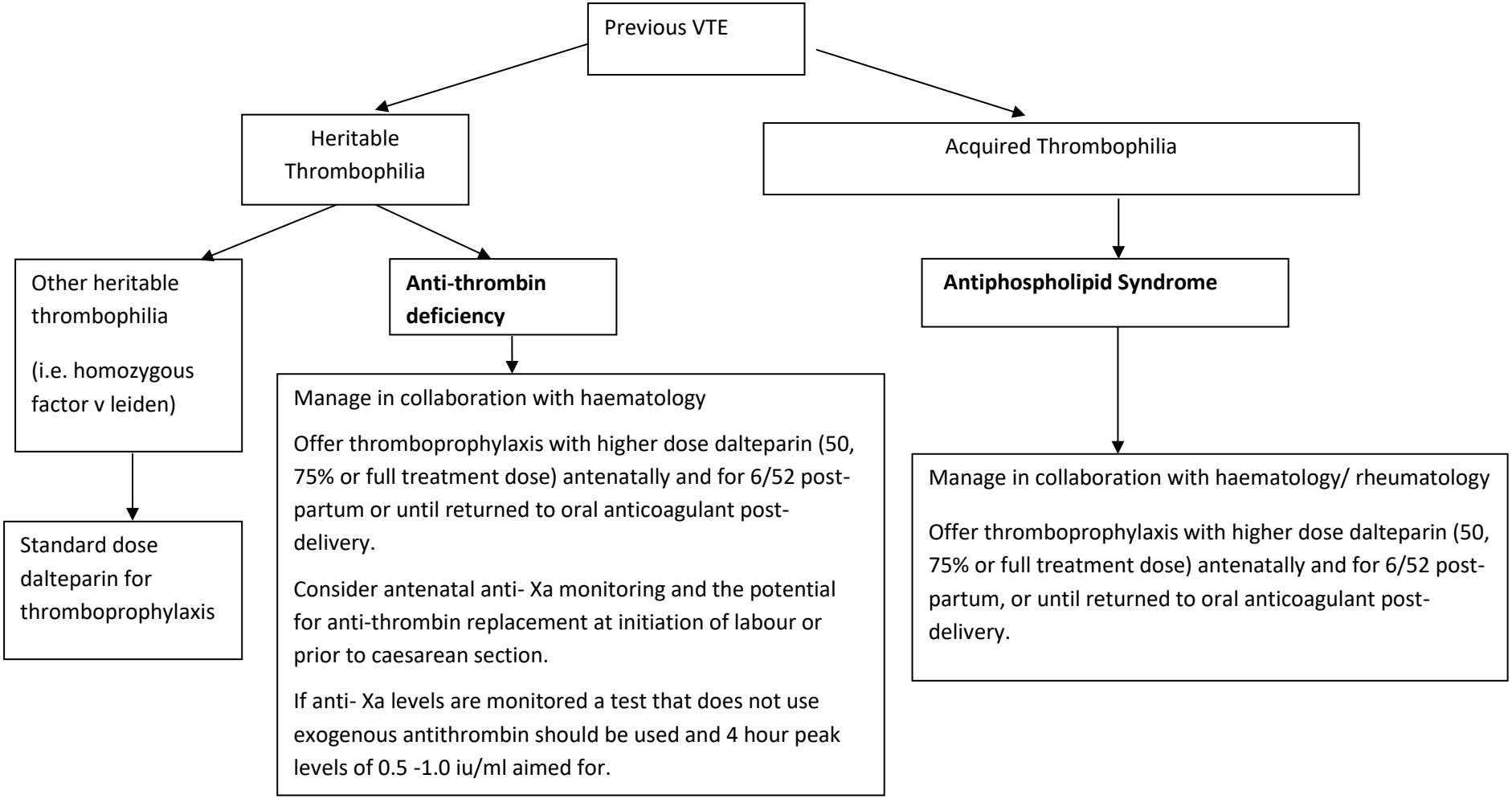
Enoxaparin is our LMWH of choice

Weight (Kg)	Standard Enoxaparin Dose	Standard Dalteparin Dose
<50	20mg daily	2500 units daily, S/C
50-90	40mg daily	5000 units daily, S/C
91-130	60mg daily *	7500 units daily, S/C
131-170	80mg daily *	10,000 units daily S/C (given in 2 divided doses) Hence 5000iu BD (0800 &2000)
>170	0.6mg/kg/day *	75 units/kg/day (given in 2 divided doses)
High prophylactic dose for 50-90kg	40mg twice daily	5000iu twice daily
*** may be given in 2 divided doses**		

Any woman commenced on LMWH antenatally must have a clear management plan documented in the chart including the plan for delivery and a referral should be made to the anaesthetic team.

3.0 MANAGEMENT OF WOMEN WITH THROMBOPHILIA ASSOCIATED VTE

Some women with previous VTE associated with acquired or inherited thrombophilia will require higher dose LMWH.



3.1 Recurrent VTE

Women with recurrent VTE may require high dose thromboprophylaxis and should be managed in collaboration with haematologists. These women should be counselled pre-conceptually to ensure they have a robust plan.

Women on long term warfarin or other oral anticoagulants should be counselled about the risks of these agents to the foetus and advised to stop their oral anticoagulation and switch to LMWH as soon as pregnancy is confirmed. Ideally within 2 weeks of the missed period and before the sixth week of pregnancy.

Women not on warfarin or other oral anticoagulants should be advised to start LMWH **as soon as they have a positive pregnancy test**. This can be organised through their GP.

3.2 Single previous VTE related to major surgery

Consider starting antenatal thromboprophylaxis from 28 weeks if no additional risk factors are present.

3.3 Testing for thrombophilia in women with previous VTE

Prior to testing for thrombophilia, women should be counselled regarding the implications for themselves and family members of a positive or negative result. The results should be interpreted by clinicians with specific expertise in the area.

Women with a family history of VTE and either antithrombin deficiency or where the specific thrombophilia has not been detected should be offered testing for antithrombin deficiency.

Women with unprovoked VTE should be offered testing for the presence of antiphospholipid antibodies.

3.4 Thrombophilia and no previous VTE

These patients should be stratified according to level of risk associated with their thrombophilia and presence of absence of a family history or other risk factors.

Women with asymptomatic antithrombin, protein or S deficiency or those with more than one thrombophilic defect (including homozygous factor V Leiden, homozygous prothrombin gene mutation and compound heterozygotes) should be referred to a local expert and antenatal prophylaxis considered. They should be recommended for six weeks postnatal thromboprophylaxis even in the absence of additional risk factors.

Heterozygosity for factor V Leiden or prothrombin gene mutation or antiphospholipid antibodies are considered as risk factors for thrombosis in asymptomatic women. In the presence of three other risk factors such women may be considered for antenatal thromboprophylaxis, if there are two other risk factors thromboprophylaxis should be considered from 28 weeks and if there is one other risk factor postnatal thromboprophylaxis for 10 days should be considered (see chart on page 6).

Women with no personal history or risk factors for VTE but who have a family history of an unprovoked or oestrogen provoked VTE in a first degree relative when aged under 50 years should be considered for thrombophilia test (do not test for protein C or S deficiency in pregnancy).

4.0 LABOUR AND DELIVERY

4.1 Intra partum assessment of risk and management of those on antenatal thromboprophylaxis

VTE risk assessment of women should be repeated and documented intrapartum.

Woman commenced on LMWH antenatally should have a clear management plan documented in the pregnancy chart including the plan for delivery.

Women receiving antenatal LMWH should be advised that, if they have any vaginal bleeding or once labour begins, they should not inject any further LMWH. They should be reassessed on admission to hospital and further doses should be prescribed by medical staff after regional anaesthesia has been discussed.

4.2 Timing of doses of antenatal thromboprophylaxis for planned deliveries.

4.2.1 **Induction of labour**

Omit dose on day of induction. Ensure mechanical thromboprophylaxis is applied

4.2.2 **Elective caesarean section**

Women should receive a **thromboprophylactic dose** of LMWH on the day prior to delivery (before 1800hrs) but omit on the morning of surgery.

Those on **treatment dose** LMWH should reduce this to prophylactic dose on the day prior to surgery and omit the dose on the morning of surgery.

Where caesarean section is undertaken in women receiving therapeutic doses of LMWH, wound drains should be considered and the skin closed with interrupted sutures to allow drainage of any haematoma.

4.3 Use of Regional anaesthesia in patients on thromboprophylaxis or anticoagulant treatment.

4.3.1 Women receiving prophylactic **(low) dose** thromboprophylactic LMWH

Wait **12 hours** after last dose before spinal / epidural catheter insertion or removal.

Allow **4 hours** after insertion / removal of catheter before next dose.

If bloody, tap on epidural insertion omit the next dose.

4.3.2 Women receiving treatment **(high) dose** LMWH or thromboprophylactic (low) dose LMWH combined with antiplatelet therapy (e.g. Aspirin in Antiphospholipid syndrome):

Wait **24 hours** after last dose of LMWH before spinal / epidural catheter insertion or removal

Allow **4 hours** after insertion / removal before next dose

If bloody, tap on epidural insertion omit the next dose

4.3.3 Women receiving continuous intravenous heparin by infusion:

The infusion should be stopped for 4 hours or until the APTT ratio has returned to normal prior to epidural / spinal insertion or removal

Wait 1 hour after insertion / removal before restarting infusion

Individual complex cases should be discussed with a senior anaesthetist and obstetrician

4.4 After delivery continued vigilance for signs of vertebral canal haematoma is essential. Women should be advised to seek medical help if symptoms of weakness, numbness, acute back pain, or bowel / bladder dysfunction develop over the next 48 hours Suspected vertebral canal haematoma must be referred and investigated without delay.

5.0 THIRD STAGE OF LABOUR

5.1 In all women in whom anticoagulants have been used, the third stage of labour should be managed actively with oxytocic therapy and controlled cord traction to minimise the risk of postpartum haemorrhage.

5.2 The first thromboprophylactic dose of LMWH should be given as soon as possible after delivery provided there is no postpartum haemorrhage and regional anaesthesia has not been used.

5.3 Women at high risk of haemorrhage with risk factors including antepartum haemorrhage, coagulopathy, progressive wound haematoma, suspected intra-abdominal bleeding and postpartum haemorrhage may be managed with anti-embolism Stockings (AES), foot impulse devices or intermittent pneumatic compression devices. Unfractionated heparin may also be considered.

5.4 If a woman develops a haemorrhagic problem whist on LMWH. The treatment should be stopped and expert haematological advice sought.

5.5 Thromboprophylaxis should be started or reinstituted as soon as the immediate risk of haemorrhage is reduced.

6.0 POSTNATAL RISK ASSESSMENT AND THROMBOPROPHYLAXIS

6.1 A postnatal VTE risk assessment must be carried out at least once after delivery and before discharge.

6.2 In the event of regional analgesia, LMWH should be given 4 hours post spinal anaesthetic or epidural catheter removal. In all other cases, give LMWH at the earliest opportunity post-delivery.

6.3 Arrangements should be made for LMWH prescription and administration (usually by the woman herself) in the community where necessary.

6.4 Doses of LMWH are based on weight. For thromboprophylaxis the booking or most recent weight can be used to guide dosing. **Enoxaparin is our LMWH of choice**

Weight (Kg)	Standard Enoxaparin Dose	Standard Dalteparin Dose
<50	20mg daily	2500 units daily, S/C
50-90	40mg daily	5000 units daily, S/C
91-130	60mg daily *	7500 units daily, S/C
131-170	80mg daily *	10,000 units daily S/C (given in 2 divided doses) Hence 5000iu BD (0800 &2000)
>170	0.6mg/kg/day *	75 units/kg/day (given in 2 divided doses)
High prophylactic dose for 50-90kg	40mg twice daily	5000iu twice daily
*** may be given in 2 divided doses**		

6.5 It is only necessary to monitor platelet count if the woman has had prior exposure to unfractionated heparin.

6.6 Doses of LMWH **should be halved in women with GFR of <30**

6.7 LMWH is safe in breast feeding.

6.8 The use of anti-embolism stockings is recommended in pregnancy and the puerperium for women who are hospitalised, and those who have a contraindication to LMWH. These include women who are hospitalised post caesarean section (combined with LMWH) and those considered to be high risk of VTE (e.g. previous VTE, more than 4 risk factors antenatally or more than 2 risk factors postnatally) and women travelling long distance for more than 4 hours.

6.9 All women with class 3 obesity (BMI > 40kg/$m^{2)}$ should be considered for prophylactic LMWH in doses appropriate for their weight (usually divided doses) for 10 days after delivery.

6.10 All women with 2 or more persisting risk factors listed in flow chart 6.5 and in postnatal VTE assessment tab should be considered for prophylactic dose LMWH appropriate for their weight for 10 days post-delivery.

6.11 All women with a previous history of confirmed VTE should be offered thromboprophylaxis with LMWH or warfarin for at least **6 weeks** postpartum regardless of the mode of delivery.

6.12 Women with thrombophilia without previous VTE should be stratified according to both the level of risk associated with their thrombophilia and the presence or absence of a family history or other risk factors.

6.13 Women with a family history of VTE and an identified thrombophilia should be considered for 6 weeks postnatal thromboprophylaxis.

6.14 All women undergoing caesarean section should be considered for thromboprophylaxis with LMWH for 10 days after delivery.

6.15 Postnatal thromboprophylaxis (adapted from the RCOG guideline)

Any previous VTE

Anyone requiring antenatal dalteparin

High risk thrombophilia *(*Antiphospholipid syndrome/ antithrombin deficiency/ protein s or c deficiency/compound or homozygous for low risk thrombophilias i.e. factor v leiden)

Low risk thrombophilia* + family hx (*Heterozygous for factor V Leiden or prothrombin G20210A mutations)

→ **High Risk**

At least 6 weeks postnatal prophylactic LMWH

Caesarean section (elective or emergency)

Any surgical procedure in the puerperium- except immediate repair of perineum

Body Mass Index (BMI) >40kg/m^2

Readmission or prolonged hospital admission for medical reasons (≥3 days) in the puerperium

MEDICAL COMORBIDITIES, e.g. Heart failure, active SLE, cancer, inflammatory bowel disease or inflammatory polyarthropathy, nephrotic syndrome, sickle cell disease, current IVDU.

→ **Intermediate risk**

At least **10 days** postnatal prophylactic LMWH

Note if persisting or > 3 risk factors, consider extending thromboprophylaxis with LMWH for 6/52 or until additional risk factors are no longer present.

Age >35 years
BMI 30 - 39
BMI >40 (= 2 risk factors)
Parity ≥3
Smoker
Family history of VTE
Low risk thrombophilia
Gross varicose veins
Current systemic infection
Immobility e.g. paraplegia, SPD, long distance travel
Current pre- eclampsia
Multiple pregnancy
Mid cavity rotational operative delivery
Preterm delivery in this pregnancy (<37/40)
Stillbirth this pregnancy
Prolonged labour (>24 hours)

→ **2** or more risk factors

→ **< 2** risk factors

→ **Lower Risk**

Mobilisation and avoidance of dehydration.

6.16 In women at a very high risk of thrombosis, unfractionated heparin may be used peripartum in preference to LMWH where there is an increased risk of haemorrhage or where regional anaesthetic techniques may be required.

If unfractionated heparin is used platelet count should be monitored every 2-3 days from day 4-14 or until the heparin is stopped.

7.0 DIAGNOSIS AND TREATMENT OF SUSPECTED VTE

7.1 There should be a high index of suspicion for VTE in pregnant women who present with symptoms and signs consistent with possible VTE, particularly if there are other risk factors.

7.2 VTE may occur in the legs (deep vein thrombosis (DVT)) lung (pulmonary embolism (PE)) or in other venous systems (such as ovarian vein).

7.3 The symptoms of VTE include:

- leg pain redness and swelling (usually unilateral)
- lower abdominal pain
- low-grade pyrexia
- dyspnoea
- chest pain
- cough/ haemoptysis
- collapse

7.4 Signs of PE:

- tachycardia
- tachypnoea
- raised JVP
- pleural rub
- loud second heart sound
- right ventricular heave

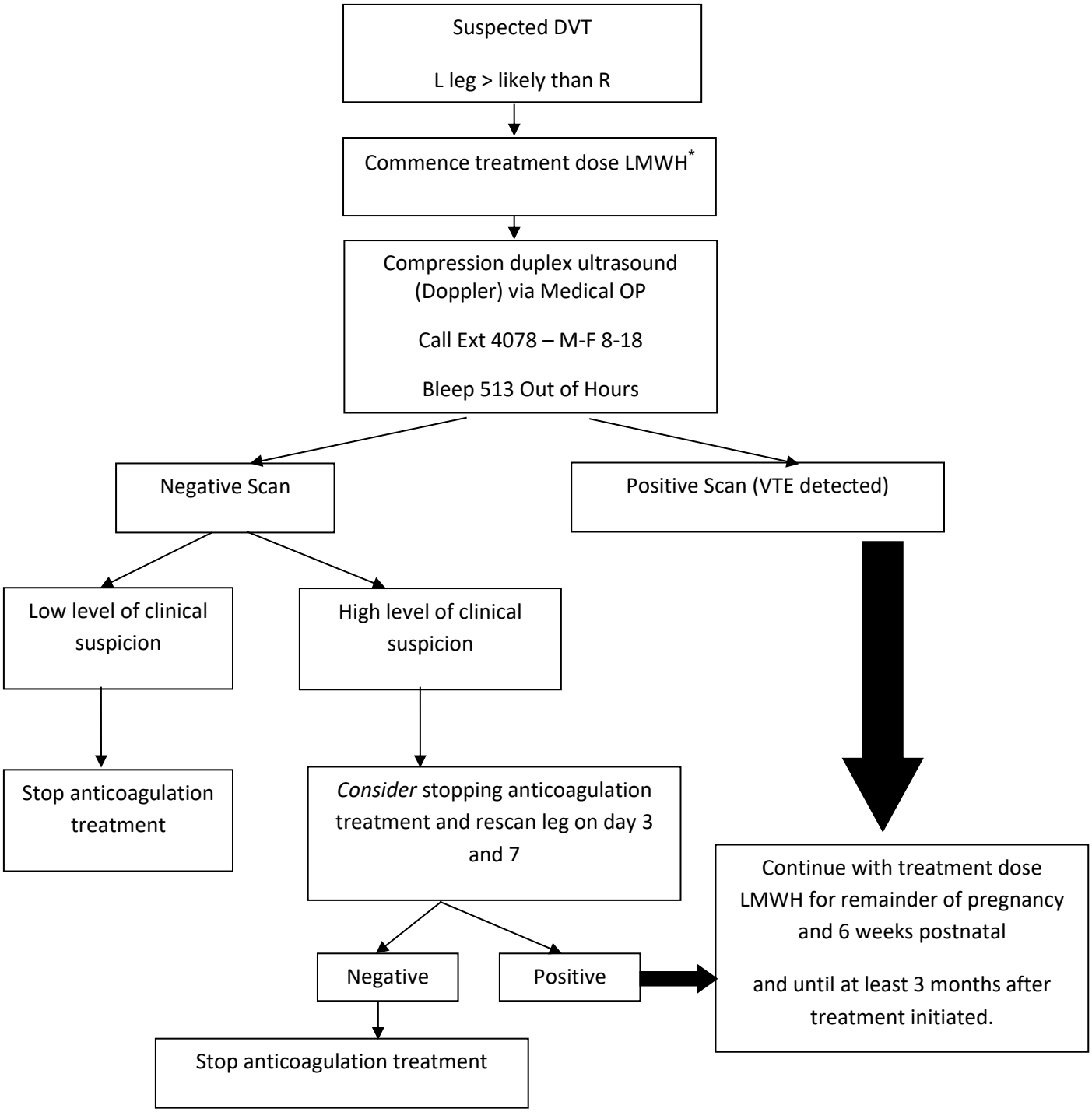
7.5 Pregnant women with symptoms of DVT who are clinically stable can be referred to Triage. If they have been seen by their GP then they can be treated and referred to the DVT clinic via primary care – see point 7.7.

7.6 Pregnant women who present with chest pain or shortness of breath should be discussed with Triage who will direct their referral to ED to allow prompt resuscitation, investigation and management as required.

7.7 Any woman with signs and symptoms suggestive of VTE should have objective testing performed expeditiously and commence treatment with LMWH until the diagnosis is excluded by objective testing, unless treatment is strongly contraindicated.

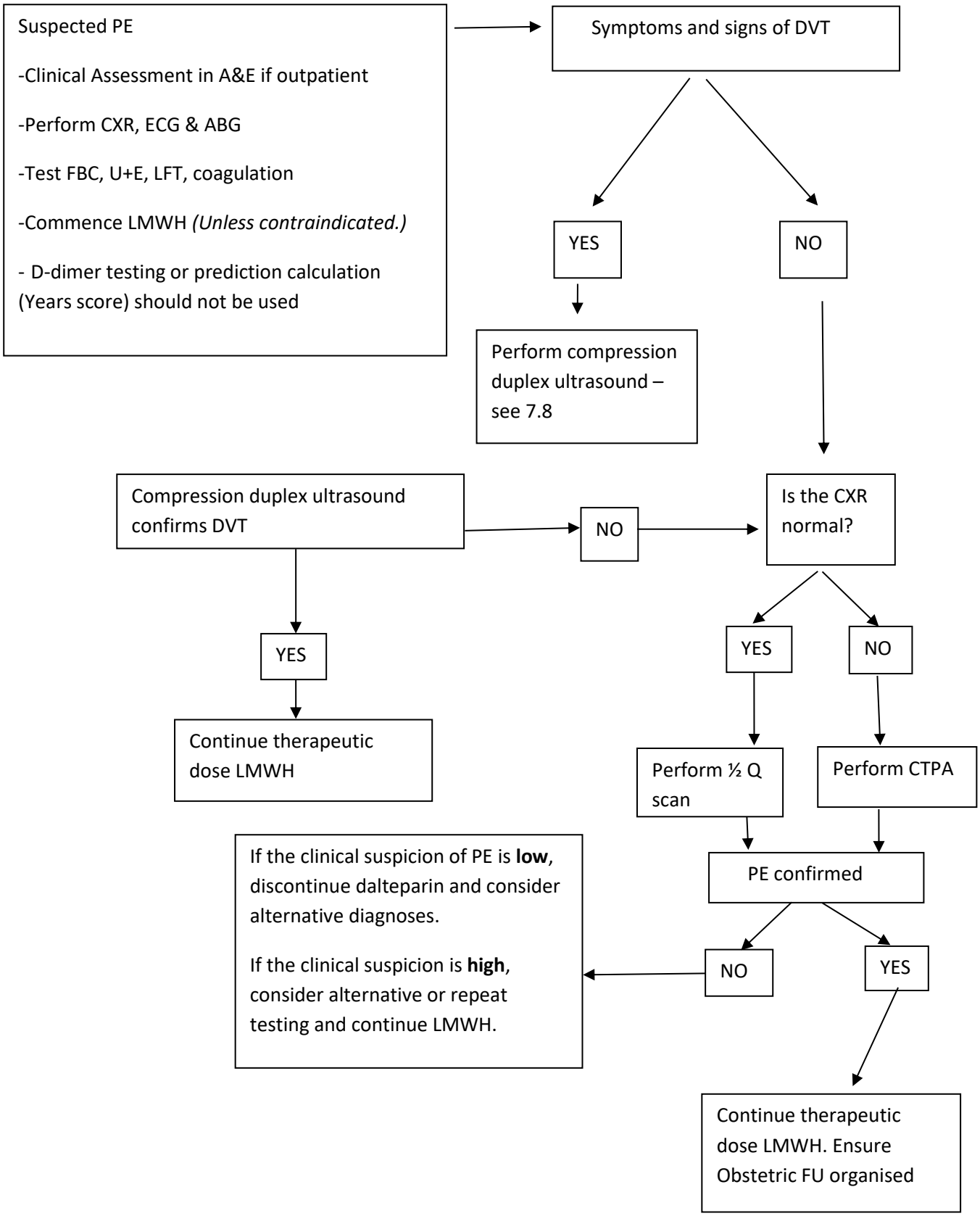
7.8 Investigations for suspected DVT

D-dimer testing or prediction calculation (for example Years score assessment) should not be performed to diagnose acute VTE in pregnancy in pregnancy



* Unless serious contraindication

7.9 Investigation of suspected pulmonary embolism in pregnancy



7.10 Facts for patient (allow patient choice where possible)

V/Q scanning carries a slightly increased risk of childhood cancer compared with CTPA (1/280,000 versus less than 1/1,000,000) but carries a lower risk of maternal breast cancer (lifetime risk increased by 13.6% with CTPA, background risk of 1/200 for study population). In both situations the absolute risk is very small.

7.11 When iliac vein thrombosis is suspected (back pain and swelling of the entire limb), magnetic resonance venography or conventional contrast venography (with abdominal shielding) may be considered.

7.12 Massive Pulmonary Embolism

Women who present with massive life threatening Pulmonary Embolism (PE) should be reviewed urgently by the on call anaesthetist and Obstetric Consultant. An urgent CTPA and portable echocardiogram should be arranged within 1 hour of presentation and discussion with the medical on call team should take place.

The decision should be made on an individual basis whether a woman receives IV unfractionated heparin, thrombolytic therapy or thoracotomy and surgical embolectomy.

Management should involve a multidisciplinary team including senior physicians, obstetricians and radiologists.

If massive PE is confirmed, or in extreme circumstances prior to confirmation, immediate thrombolysis should be considered.

Intravenous unfractionated heparin is the preferred treatment in massive PE with cardiovascular compromise (hyperlink heparin guideline on Ian) and must be prescribed under the direction of senior clinicians.

7.13 Management of VTE in pregnancy

LMWH is the treatment of choice for VTE in pregnancy and the immediate postnatal period (except in cases of massive PE with cardiovascular compromise.)

Once VTE has been diagnosed in pregnancy, treatment with therapeutic subcutaneous LMWH should be continued for the remainder of the pregnancy.

Treatment dose LMWH is based on **current weight**. **Enoxaparin** is our LMWH of choice

Weight Kg	Treatment dose Enoxaparin	Treatment dose of Fragmin
<50	40mg bd or 60mg od	5000 iu S/C bd or 10000iu od
50-69	60mg bd or 90mg od	6000 iu S/C bd or 12000iu od
70-89	80mg bd or 120mg od	8000 iu S/C bd or 16000iu od
90- 109	100mg bd or 150mg od	10000 iu S/C bd or 20000iu od
110-125	120mg bd 0r 180mg od	12000 iu S/C bd or 24000iu od
>125	Discuss with Hematologist	Discuss with Hematologist

Women who have been commenced on therapeutic anticoagulant therapy for a VTE in pregnancy should be continued for at least 6 weeks postnatally and until at least 3 months of treatment has been given in total.

In the initial management of DVT the leg should be elevated and graduated elastic compression stockings applied to reduce pain and swelling. Mobilisation with compression stockings should be encouraged.

Because of their adverse effect on the foetus, vitamin K antagonists, such as warfarin should not be used for antenatal VTE treatment.

Women should be offered a choice of LMWH or oral anticoagulation for postnatal therapy after discussion about the need for regular blood tests and monitoring of warfarin, particularly in the first 10 days of treatment.

Postpartum warfarin should be avoided until at least the fifth day and for longer in women at increased risk of postpartum haemorrhage.

Neither warfarin nor LMWH are contraindicated in breast feeding.

Women should be advised that prolonged use of LMWH (>12 weeks) is associated with a significantly lower chance of developing post-thrombotic syndrome.

Routine platelet count should not be carried out (unless unfractionated heparin has been given).

Pregnant women who develop heparin induced thrombocytopenia or have a heparin allergy and require continuing anti-coagulant therapy should be managed with an alternative anticoagulant under specialist advice.

8.0 POSTNATAL FOLLOW UP FOR PATIENTS WITH VTE IN PREGNANCY AND THE PUPERIUM

8.1 Women who develop VTE during pregnancy or the puerperium should be reviewed in the antenatal clinic and include liaison with the haematology team.

8.2 At the postnatal review, an assessment should be made of post-thrombotic venous damage, thrombophilia tests should be reviewed and arrangements made to repeat them if necessary. Advice should be given on the need for thromboprophylaxis in any future pregnancy and at other times of increased risk. Hormonal contraception should be discussed.

8.3 Women should be taught to self-inject LMWH. Arrangements should be made to allow safe disposal of the needles and syringes.

8.4 Outpatient follow up should include clinical assessment and advice with assessment of peak anti –Xa levels if appropriate (extremes of body weight (<50kg, >90kg), renal failure etc.).

9.0 MONITORING COMPLIANCE WITH THIS GUIDELINE

9.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per the Trust Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

9.2 Relevant Policies:

- Incident reporting policy and procedure
- Claims management policy and procedure
- Policy and Procedure for the Management of Complaints, Concerns, Comments and Compliments

10. REFERENCES

10.1 MBRRACE-UK: Mothers and Babies: Reducing Risk through Audits and Confidential Enquires across the UK. Saving Lives, Improving Mothers' Care. Lessons learned to inform future maternity care from the UK and Ireland Confidential Enquires into Maternal Deaths and Morbidity 2018-2020. Published November 2022.

10.1.1 Reducing the Risk of Venous Thromboembolism during Pregnancy and the Puerperium. Green-top Guideline No. 37a. April 2015
Thromboembolic Disease in Pregnancy and the Puerperium: Acute Management. Green-top Guideline No. 37b. April 2015

11. PUBLICATION DETAILS

Author of Clinical Guideline	Labour Ward Lead Consultant
Division/ Department responsible for Clinical Guideline	Specialist Services/CWH/Maternity
Contact details	6682
Version number	2.6 (addition of Enoxaparin)
Replaces version number	2.5
Date written	September 2004
Approving body and date approved	Maternity Governance Forum Chairs action 14/03/2025 (drug update)
Review date	03/12/2025
Expiry date	03/05/2026
Date document becomes live	04/05/2023